

The Oura Ring Versus Medical-Grade Sleep Studies: A Systematic Review and Meta-Analysis

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Abstract

Objective. To evaluate the validity of the Oura Ring (OR; Oura Health Ltd.) in measuring sleep parameters compared to medical-grade sleep studies including polysomnography (PSG) or actigraphy (ACT).

Data Sources. PubMed, Scopus, and CINAHL.

Review Methods. A systematic review and meta-analysis was conducted in accordance with PRISMA guidelines. Studies were included if they evaluated sleep parameters measured simultaneously by the OR and PSG or ACT. Outcomes assessed included Total Sleep Time (TST), Sleep Efficiency (SE), Wake After Sleep Onset (WASO), Sleep Onset Latency (SOL), Light Sleep Time (LST), Deep Sleep Time (DST), and Rapid Eye Movement (REM) sleep time. Mean differences with 95% confidence intervals were calculated using a random-effects model. A $P < .05$ was considered statistically significant.

Results. Out of 2104 articles, 6 studies ($n = 388$) were included. There were no statistically significant differences between the OR and PSG/ACT for TST (MD: -2.97 min; 95% confidence interval [CI]: -10.27 to 4.33), SE (MD: -1.32% ; 95% CI: -2.76 to 0.12), WASO (MD: 1.64 min; 95% CI: -12.57 to 15.86), SOL (MD: 0.48 min; 95% CI: -2.93 to 3.89), LST (MD: -4.27 min; 95% CI: -24.68 to 16.13), DST (MD: 1.39 min; 95% CI: -10.45 to 13.23), and REM sleep time (MD: -3.89 min; 95% CI: -17.23 to 9.46).

Conclusion. The OR demonstrates comparable accuracy to PSG and ACT for commonly measured sleep parameters, supporting its utility as a self-monitoring tool. This could prompt earlier clinical evaluation in symptomatic individuals or support remote monitoring of sleep.

Keywords

actigraphy, diagnostic accuracy, meta-analysis, Oura Ring, polysomnography, sleep monitoring, sleep tracking, wearable devices

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Wearable devices that monitor real-time health metrics have gained popularity in recent years, driven by consumer interest in health monitoring, preventive health, and fitness tracking. These devices offer individuals greater autonomy over their health. Among these devices, the Oura Ring (OR; Oura Health Ltd.) has gained significant traction due to its compact design and ability to track a range of physiologic parameters. It has increasingly been evaluated in research contexts as a self-monitoring health tool tracking heart rate variability, body temperature, physical activity, and of our interest, sleep patterns.^{1,2} The OR estimates sleep stages and efficiency using infrared photoplethysmography sensors, body temperature sensors, and an accelerometer. By analyzing patterns in heart rate variability, body temperature, and movement, the OR's proprietary algorithms classify sleep into light, deep, and REM stages and calculate metrics such as sleep onset latency and sleep efficiency.

Accurate assessment of sleep is crucial as certain sleep cycle patterns are linked to sleep disorders such as sleep apnea, narcolepsy, and insomnia, amongst others.³⁻⁵ Sleep quality and duration are tightly linked to cognitive function, metabolic health, and cardiovascular disease risk.⁶⁻⁸ Traditional methods such as polysomnography (PSG) or actigraphy (ACT) remain the medical-grade standard for sleep evaluation, but are limited by their high cost and inconvenience. As a result, there has been

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growing interest in portable, consumer-accessible devices that can reliably assess sleep outside of laboratory settings.

Validation studies have shown that while total sleep time estimates are highly accurate in the OR compared to PSG, stage classification has moderate accuracy.^{2,9,10} Given the growing use of wearable devices in personal health management, it is critical to assess their validity and reliability. The purpose of this systematic review and meta-analysis is to evaluate the performance of the OR in measuring sleep parameters compared to gold-standard methods, namely PSG and ACT, to determine its potential utility as a reliable self-monitoring tool and its limitations within clinical application.

Methods

Search Criteria

The study adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

guidelines (**Figure 1**).¹¹ comprehensive literature search was performed across PubMed (National Library of Medicine, National Institutes of Health), Scopus (Elsevier), and CINAHL (EBSCO) from their inception through February 5, 2025. The PubMed search strategy incorporated both subject headings (eg, MeSH terms) and relevant keywords related to wearable devices and PSG or ACT. This strategy was adapted for Scopus and CINAHL by substituting MeSH terms with each database's specific subject headings. A detailed account of the search criteria and the number of results retrieved from each database is provided in **Table 1**. All references were imported into Covidence (Veritas Health Innovation Ltd.) for screening and review.

Selection Criteria

Studies involving patients who have undergone simultaneous PSG or ACT and sleep tracking with the OR were included. Studies were included if both medical-grade sleep

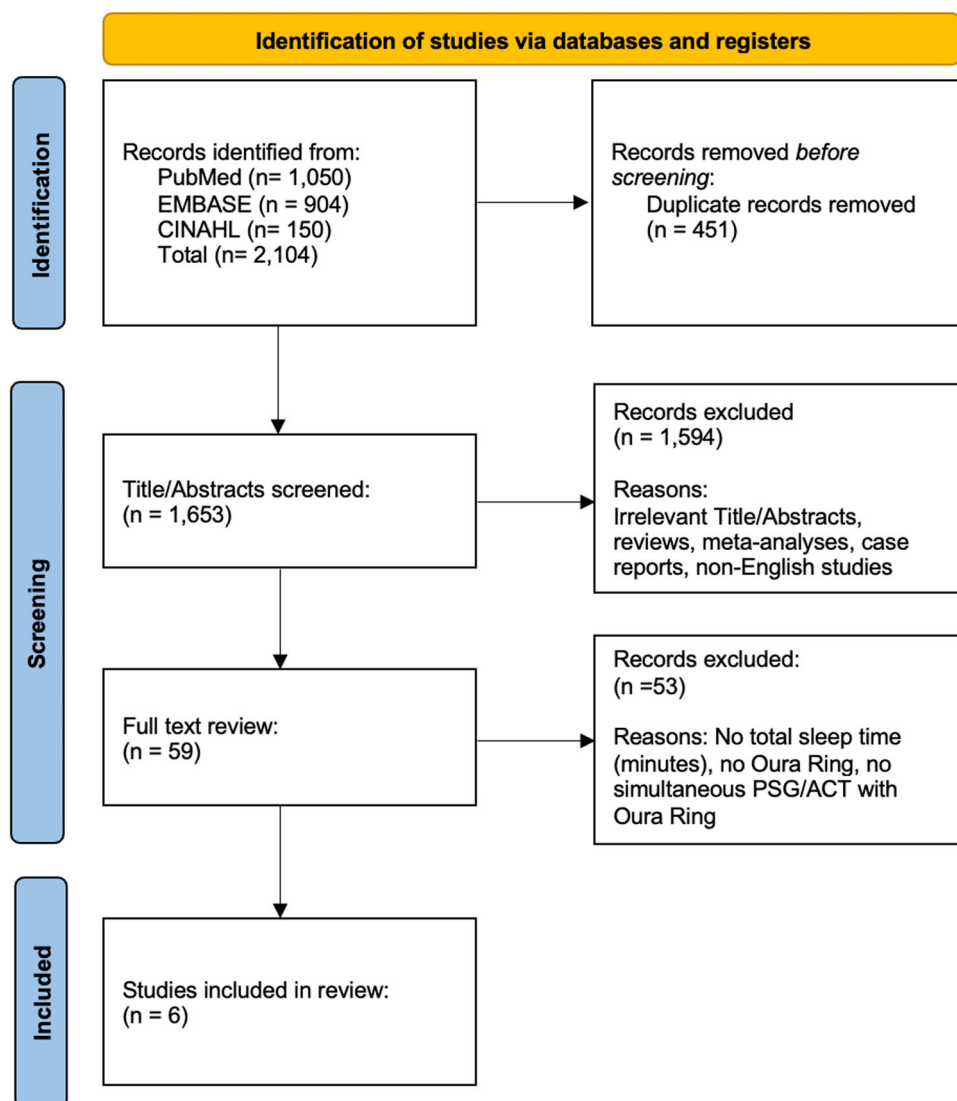


Figure 1. PRISMA flow diagram of the literature review. PRISMA, preferred reporting items for systematic reviews and meta-analyses.

tracking devices and OR recorded TST at the minimum. Studies were excluded if sleep outcomes were not simultaneously measured by medical-grade sleep trackers and the OR. Two reviewers (SK and AFI) first independently reviewed titles and abstracts to identify articles that met the inclusion criteria. Disagreements were resolved by way of discussion with the senior author (MMC). Both reviewers subsequently reviewed full texts independently to identify articles that met the inclusion criteria with disagreements after discussion with the senior author.

Critical Appraisal and Quality Assessment

Each study's level of evidence was assessed using the Oxford Centre for Evidence-Based Medicine (OCEBM) criteria (Table 2).¹⁸ Risk of bias was assessed for each included study using the QUADAS-C tool. In systematic reviews of diagnostic test accuracy, the QUADAS-C tool is used to assess the risk of bias in comparative accuracy studies that evaluate 2 or more index tests. The QUADAS-C tool evaluates four domains relevant to comparative diagnostic accuracy studies: patient selection, index test conduct and interpretation, reference standard conduct and interpretation, and flow and timing (Figure 2).¹⁹ For each domain, risk of bias was rated as “low,” “some concerns,” or “high.” An overall risk of bias judgment was assigned based on the highest level of risk identified across any domain.

Applicability concerns were also considered but are reported separately. All assessments were conducted following the QUADAS-C guidance criteria.

Data Extraction

Data were extracted independently by 2 reviewers and compared for accuracy. Disagreements were resolved by discussion. The Total Sleep Time (TST), Sleep Efficacy (SE), Wake After Sleep Onset (WASO), Sleep Onset Latency (SOL), Light Sleep, Deep Sleep, and REM Sleep were extracted from medical-grade sleep trackers, either polysomnography (PSG) or ACT, and the OR.

Statistical Analysis

All statistical analyses were performed using Comprehensive Meta-Analysis (CMA) software, version 4. For studies comparing outcomes between OR and PSG, mean differences (MDs) and corresponding 95% confidence intervals (CIs) were calculated. Given the anticipated heterogeneity across studies, a random-effects meta-analysis model was employed. To assess the impact of a single study on the overall outcome, a sensitivity analysis was conducted using the leave-one-out method, whereby the meta-analysis was repeated after sequentially removing each study. Publication bias was evaluated through visual inspection of funnel plots

Table 1. Search Strategies and Results

Databases	Combination terms	Number of references obtained
PubMed	(sleep tracker OR wearable device OR ring OR watch) AND (Polysomnography OR actigraphy)	1050
EMBASE/SCOPUS	(“sleep tracker” OR “wearable device” OR “ring” OR “watch”) AND (“polysomnography” OR “actigraphy”)	904
EBSCO/CINAHL	(sleep tracker OR wearable device OR ring OR watch) AND (Polysomnography OR actigraphy)	150

Table 2. Summary of Study Characteristics and Patient Demographics for Studies Included in the Systematic Review and Meta-Analysis

Author	Sample size (n)	Country	LOE	Meanage, years (SD)	Sex (%)	Sleep study	Outcomes measured
Altini 2021 ¹²	118	Singapore, Finland, and USA	2	29.7 (16.7)	M: 44.9%, F: 55.1%	PSG	TST, LST, DST, REM
AsgariMehrabadi 2020 ¹³	45	Finland	3	33.1 (6.4)	M: 48.9%, F: 51.1%	ACT	TST, SE, WASO
deZambotti 2019 ¹⁴	41	USA	2	17.2 (2.4)	M: 68.3%, F: 31.7%	PSG	TST, LST, DST, REM, SOL, WASO
Kainec 2024 ¹⁵	53	USA	2	22.5 (3.5)	M: 41.5%, F: 58.5%	PSG	TST, LST, DST, REM, WASO
Robbins 2024 ¹⁶	35	USA	2	N/A	M: 42.9%, F: 57.1%	PSG	TST, LST, DST, REM, SE, SOL, WASO
Svensson 2024 ¹⁷	96	Japan	2	41.9 (13.8)	M: 49.0%, F: 51.0%	PSG	TST, LST, DST, REM, SE, SOL, WASO

Abbreviations: ACT, actigraphy; DST, deep sleep time; LOE, level of evidence; LST, light sleep time; N/A, no data; PSG, polysomnography; REM, rapid eye movement sleep time; SD, standard deviation; SE, sleep efficiency; SOL, sleep onset latency; TST, total sleep time; WASO, wake after sleep onset

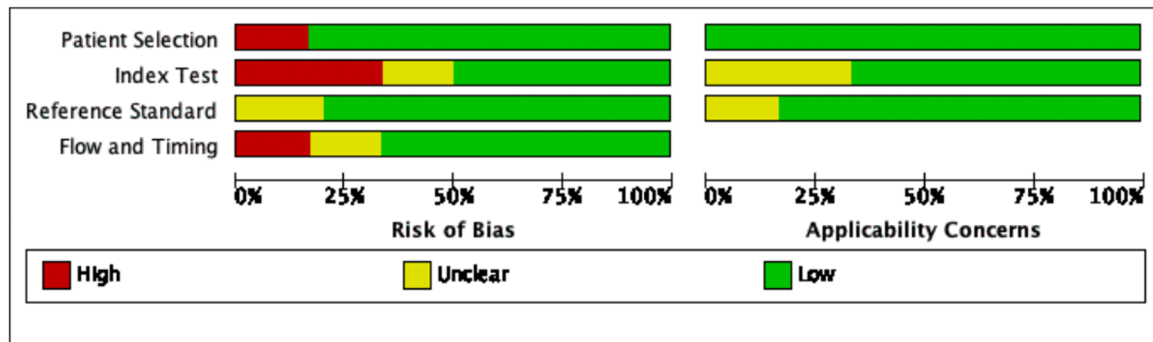


Figure 2. Risk of bias and applicability concerns graph: review authors' judgements about each domain presented as percentages across included studies.

and Egger's regression test. All tests were 2-sided, and a $P < .05$ was considered statistically significant.

Results

Search Results and Study Characteristics

The initial literature search yielded 2104 articles. Following duplicates removal, 1653 articles were included for screening. After title/abstract screening, 1594 records were excluded. Full-text review screening of the remaining 59 articles was conducted, which yielded 6 final studies included in the meta-analysis.^{12–17} A PRISMA diagram detailing the literature search is shown in **Figure 1**. The included studies were published between 2019 and 2024, with the majority of the studies from the United States (50%). The Oxford Level of Evidence for each study was assessed and categorized as either Level 2 or 3, as detailed in **Table 2**. Quality assessment was performed using the QUADAS-C tool for all 6 included studies (**Figure 2**).

Patient Characteristics

In the quantitative analysis, 388 patients with a mean (SD) age of 64.7 (9.5) years were included to assess outcomes between OR and PSG. Outcomes measured include TST, SE, WASO, SOL, Light Sleep, Deep Sleep, and REM Sleep. The author, year, sample size, level of evidence, demographics (age and gender), and sleep outcomes were extracted for each included study (**Table 2**). All included studies evaluated generally healthy participants, with 5 of the 6 explicitly confirming the absence of sleep disorders through either self-report, validated instruments, or polysomnography.

Meta-Analysis of OR Versus PSG

Across 6 studies comparing the OR to PSG, there was no significant difference in TST, with a mean difference of -2.97 minutes (95% CI: -10.27 to 4.33 ; $P = .42$) (**Figure 3**). Analysis of LST showed a non-significant mean difference of -4.27 minutes (95% CI: -24.68 to 16.13 ; $P = .68$). For DST, the OR measurements differed from PSG by an

average of 1.39 minutes (95% CI: -10.45 to 13.23 ; $P = .82$), which was not statistically significant. SE time was slightly lower in the OR recordings compared to PSG, with a mean difference of -1.32% (95% CI: -2.76 to 0.12 ; $P = .07$), but this also did not reach significance (**Figure 3**). SOL differed by only 0.48 minutes between devices (95% CI: -2.93 to 3.89 ; $P = .78$), again showing no meaningful difference. WASO showed a small, non-significant difference of 1.64 minutes (95% CI: -12.57 to 15.86 ; $P = .82$). Lastly, REM sleep time was slightly lower with OR compared to PSG, but the difference of -3.89 minutes (95% CI: -17.23 to 9.46 ; $P = .57$) was not statistically significant.

Sensitivity Analysis

On one-study removal sensitivity analysis, there was no significant difference in the overall effects on TST, LST and DST outcomes as shown in **Figure 4**. For the SE, SOL and WASO, there were similar outcomes with no significant difference on sensitivity analysis (**Figure 4**).

Publication Bias

Egger's test for publication bias showed no significant bias for most sleep outcomes except for sleep onset latency (**Figure 5**), where a significant result was observed ($t = 20$, $P = .03$), suggesting potential publication bias for this outcome (**Table 3**).

Discussion

Sleep-related complaints are among the most common reasons for referral to otolaryngology. Although polysomnography remains the diagnostic gold standard, its high cost, limited accessibility, and patient inconvenience often delay timely evaluation. Wearable devices offer a promising way to bridge these gaps, particularly in the current era of personalized medicine and digital health. Because sleep disturbances are strongly associated with cardiometabolic disease, mood disorders, and cognitive decline, accurate home-based monitoring tools have the

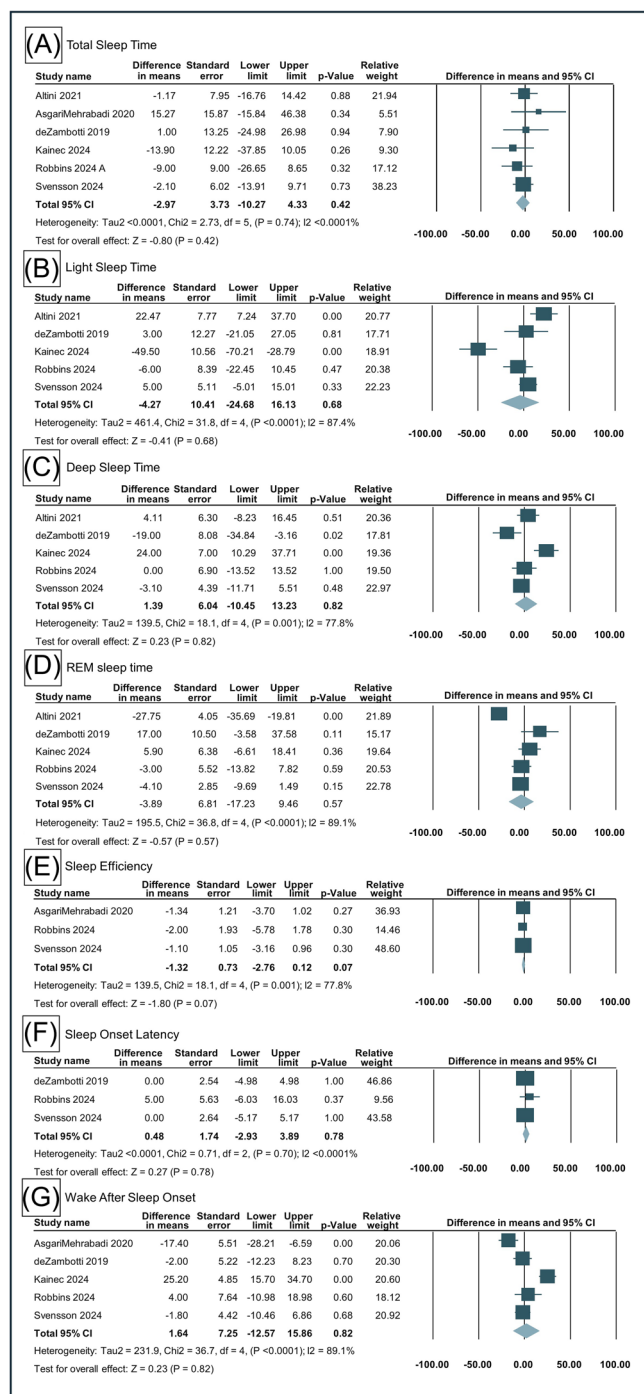


Figure 3. Forest plot illustration of sleep outcomes between OR and PSG/ACT. ACT, actigraphy; OR, Oura Ring; PSG, polysomnography.

potential to play a critical role in both screening and longitudinal management of sleep disorders.^{20–23}

Findings from this meta-analysis suggest that the Oura Ring, by demonstrating strong agreement with medical-grade sleep studies across multiple parameters, may serve as a practical adjunct in ENT practice. Notably, no statistically significant differences were observed between the OR and PSG for TST, SE, SOL, WASO, or specific sleep stages. These findings are consistent with prior



Figure 4. Sensitivity analysis with one-study removal analysis for sleep outcomes between OR and PSG/ACT. ACT, actigraphy; CI, confidence interval; OR, Oura Ring; PSG, polysomnography.

validation studies, which have shown that the OR has high sensitivity in detecting sleep versus wakefulness; however, our studies show high, rather than moderate, accuracy in distinguishing between sleep stages.^{24–26} Our sensitivity analyses further confirmed the robustness of these results, with no single study significantly altering the pooled outcomes. While Egger's test did not suggest publication bias for most outcomes, a significant result for sleep onset latency ($P = .03$) indicates potential

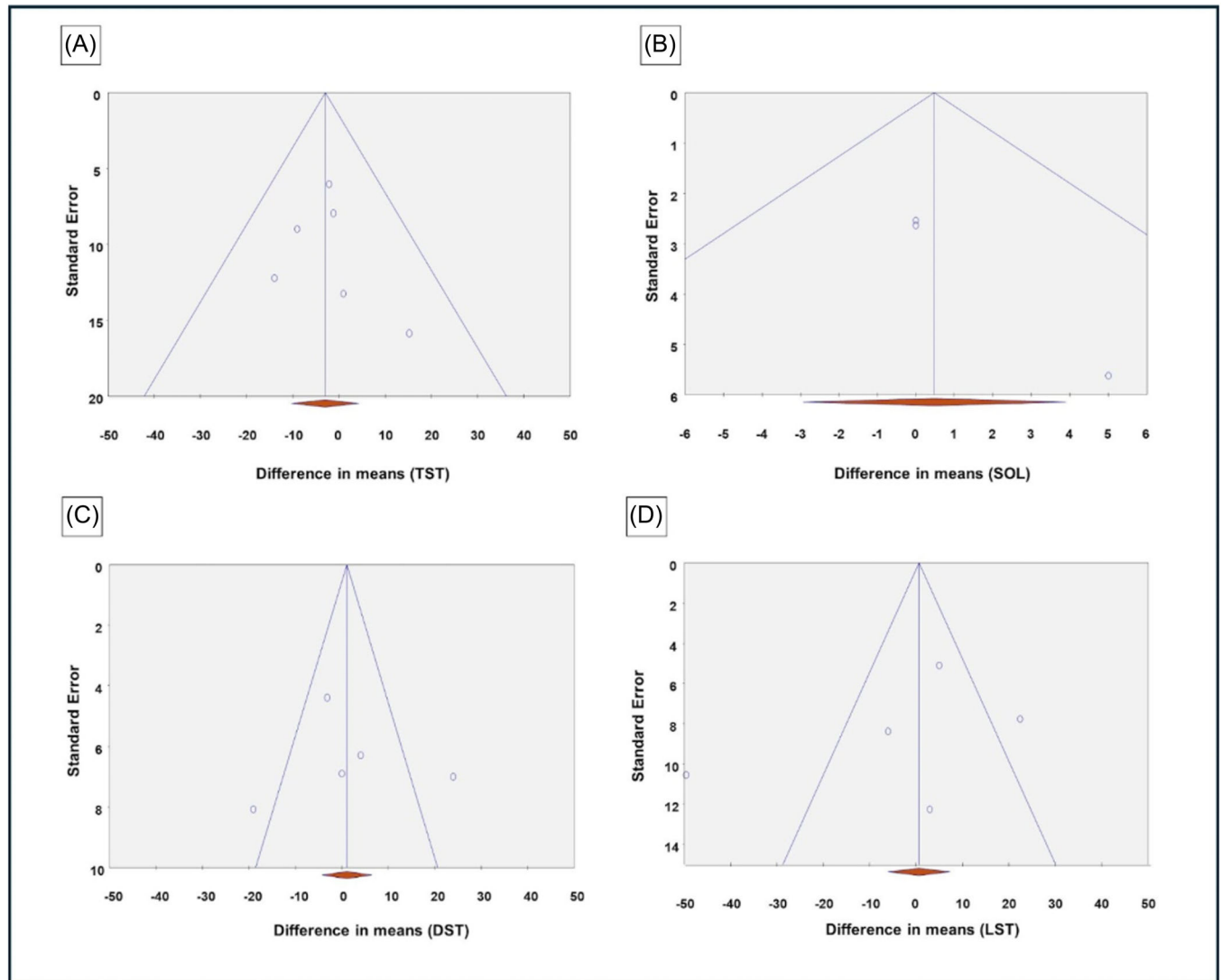


Figure 5. Funnel plot illustrating publication bias for (A) TST, (B) SOL, (C) DST, and (D) LST. DST, Deep Sleep Time; LST, Light Sleep Time; SOL, Sleep Onset Latency; TST, Total Sleep Time.

Table 3. Publication Bias Assessment for OR Versus PSG Outcomes

Outcomes	Effect size (beta) with 95% CI	t-value (Egger)	P-value (Egger)	Publication bias
Total sleep time	0.42 (CI: -2.45-3.29)	0.4	.71	Not significant
Light sleep time	-3.63 (CI: -17.3-10.0)	0.84	.46	Not significant
Deep sleep time	0.74 (CI: -15.2-16.6)	0.15	.89	Not significant
Sleep Efficiency	-1.02 (CI: -2.71-0.67)	7.65	.08	Not significant
Sleep Onset Latency	1.64 (CI: 0.60-2.68)	20	.03	Significant
Wake After Sleep Onset	-2.21 (CI: 31.4-27.0)	0.24	.82	Not significant
REM sleep time	2.54 (CI: -9.40-14.5)	0.68	.55	Not significant

Abbreviations: CI, confidence interval; OR, Oura Ring; PSG, polysomnography.

selective reporting for this parameter. This should be considered when interpreting the agreement between the OR and PSG in measuring SOL.

While it is not a replacement for PSG in the diagnosis of conditions such as obstructive sleep apnea or narcolepsy, the OR can provide continuous data on sleep

architecture and efficiency that may help flag deviations from normal patterns. This could aid in the screening and triage of patients presenting with nonspecific sleep complaints, prompting earlier clinical evaluation in symptomatic individuals. Given its accessibility and ease of use, the OR may therefore serve as a valuable screening

or adjunctive tool in sleep medicine. The OR can also help in monitoring outcomes following surgical interventions such as adenotonsillectomy or nasal airway surgery, and support longitudinal tracking in patients. For instance, trends in reduced REM or deep sleep, elevated WASO, or decreased sleep efficiency over time may suggest underlying sleep pathology or worsening disease control.^{27–29} By providing a scalable, patient-friendly, and accessible method for home-based sleep monitoring, the OR may allow otolaryngologists to identify at-risk individuals earlier, enhance patient engagement in their care, and prioritize formal testing for those most likely to benefit—all of which are particularly valuable in clinical settings.

Several limitations must be considered. First, heterogeneity in study designs, sample populations, and sleep metrics across included studies may influence the generalizability of the findings. Most included studies evaluated healthy adults, limiting extrapolation to populations with sleep disorders, comorbidities, or advanced age. Second, while the QUADAS-C tool indicated a low to moderate risk of bias overall, variability in reference standards (PSG vs ACT) could introduce inconsistency. Finally, the evolving nature of wearable device algorithms necessitates continuous validation as technology advances.

Future research should focus on validating the OR and similar devices in more diverse populations, including individuals with sleep disorders such as obstructive sleep apnea and insomnia. Additionally, real-world longitudinal studies assessing how wearable-derived sleep metrics correlate with clinical outcomes would be valuable.

Conclusion

The OR shows strong agreement with medical-grade sleep studies across multiple sleep parameters, supporting its use as a reliable self-monitoring tool. Its potential to identify abnormal sleep patterns highlights a promising role in the early detection and monitoring of sleep-related disorders. This meta-analysis shows that OR has comparable diagnostic accuracy with traditional sleep medical-grade sleep studies. While the OR cannot yet replace clinical diagnostic testing, it offers a valuable, accessible adjunct in clinical ENT settings, research, and population health applications.

Author Contributions

Sofia Khan, substantial contributions to the conception or design of the work and the acquisition, analysis, and interpretation of data for the work, drafting the work and reviewing it critically for important intellectual content, final approval of the version to be published and agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved; **Alexa F. Ibrahim**, substantial contributions to the conception or design of the work and the acquisition, analysis, and interpretation of data for the work, drafting the work and reviewing it critically for important intellectual content, final approval of the version to be published and agreement to be

accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved; **Srivatsa Surya Vasudevan**, substantial contributions to the conception or design of the work and the acquisition, analysis, and interpretation of data for the work, drafting the work and reviewing it critically for important intellectual content, final approval of the version to be published and agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved; **Olivia E. Quatela**, substantial contributions to the conception or design of the work and the acquisition, analysis, and interpretation of data for the work, drafting the work and reviewing it critically for important intellectual content, final approval of the version to be published and agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved; **Douglas P. Nanu**, substantial contributions to the conception or design of the work and the acquisition, analysis, and interpretation of data for the work, drafting the work and reviewing it critically for important intellectual content, final approval of the version to be published and agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved; **Michele M. Carr**, substantial contributions to the conception or design of the work and the acquisition, analysis, and interpretation of data for the work, drafting the work and reviewing it critically for important intellectual content, final approval of the version to be published and agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Disclosures

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References

1. Cao R, Azimi I, Sarhaddi F, et al. Accuracy assessment of Oura Ring nocturnal heart rate and heart rate variability in comparison with electrocardiography in time and frequency domains: comprehensive analysis. *J Med Internet Res*. 2022; 24(1):e27487. doi:10.2196/27487
2. Kristiansson E, Fridolfsson J, Arvidsson D, Holmäng A, Börjesson M, Andersson-Hall U. Validation of Oura Ring energy expenditure and steps in laboratory and free-living. *BMC Med Res Methodol*. 2023;23(1):50. doi:10.1186/s12874-023-01868-x
3. Nozawa S, Urushihata K, Machida R, Hanaoka M. Sleep architecture of short sleep time in patients with obstructive sleep apnea: a retrospective single-facility study. *Sleep Breath*. 2022;26(4):1633-1640. doi:10.1007/s11325-021-02533-7
4. Zhang Y, Ren R, Yang L, et al. Polysomnographic nighttime features of narcolepsy: a systematic review and

- meta-analysis. *Sleep Med Rev.* 2021;58:101488. doi:10.1016/j.smrv.2021.101488
5. Baglioni C, Regen W, Teghen A, et al. Sleep changes in the disorder of insomnia: a meta-analysis of polysomnographic studies. *Sleep Med Rev.* 2014;18(3):195-213. doi:10.1016/j.smrv.2013.04.001
 6. Ge X, Han F, Huang Y, et al. Is obstructive sleep apnea associated with cardiovascular and all-cause mortality? *PLoS One.* 2013;8(7):e69432. doi:10.1371/journal.pone.0069432
 7. Li M, Wang N, Dupre ME. Association between the self-reported duration and quality of sleep and cognitive function among middle-aged and older adults in China. *J Affect Disord.* 2022;304:20-27. doi:10.1016/j.jad.2022.02.039
 8. Knauert M, Naik S, Gillespie MB, Kryger M. Clinical consequences and economic costs of untreated obstructive sleep apnea syndrome. *World J Otorhinolaryngol Head Neck Surg.* 2015;1(1):17-27. doi:10.1016/j.wjorl.2015.08.001
 9. Walch O, Huang Y, Forger D, Goldstein C. Sleep stage prediction with raw acceleration and photoplethysmography heart rate data derived from a consumer wearable device. *Sleep.* 2019;42(12):zsz180. doi:10.1093/sleep/zsz180
 10. Henriksen A, Svartdal F, Grimsgaard S, Hartvigsen G, Hopstock LA. Polar Vantage and Oura physical activity and sleep trackers: validation and comparison study. *JMIR Form Res.* 2022;6(5):e27248. doi:10.2196/27248
 11. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ.* 2021;372:n71. doi:10.1136/bmj.n71
 12. Altini M, Kinnunen H. The promise of sleep: a multi-sensor approach for accurate sleep stage detection using the Oura Ring. *Sensors.* 2021;21(13):4302. doi:10.3390/s21134302
 13. Asgari Mehrabadi M, Azimi I, Sarhaddi F, et al. Sleep tracking of a commercially available smart ring and smart-watch against medical-grade actigraphy in everyday settings: instrument validation study. *JMIR Mhealth Uhealth.* 2020;8(10):e20465. doi:10.2196/20465
 14. de Zambotti M, Rosas L, Colrain IM, Baker FC. The sleep of the ring: comparison of the Oura sleep tracker against polysomnography. *Behav Sleep Med.* 2019;17(2):124-136. doi:10.1080/15402002.2017.1300587
 15. Kainec KA, Caccavaro J, Barnes M, Hoff C, Berlin A, Spencer RMC. Evaluating accuracy in five commercial sleep-tracking devices compared to research-grade actigraphy and polysomnography. *Sensors.* 2024;24(2):635. doi:10.3390/s24020635
 16. Robbins R, Weaver MD, Sullivan JP, et al. Accuracy of three commercial wearable devices for sleep tracking in healthy adults. *Sensors.* 2024;24(20):6532. doi:10.3390/s240206532
 17. Svensson T, Madhawa K, Nt H, Chung U, Svensson AK. Validity and reliability of the Oura Ring Generation 3 (Gen3) with Oura sleep staging algorithm 2.0 (OSSA 2.0) when compared to multi-night ambulatory polysomnography: a validation study of 96 participants and 421,045 epochs. *Sleep Med.* 2024;115:251-263. doi:10.1016/j.sleep.2024.01.020
 18. OCEBM Levels of Evidence Working Group = Jeremy Howick ICJLL, Glasziou P, Greenhalgh T, Heneghan C, et al. "The Oxford Levels of Evidence 2". Oxford Centre for Evidence-Based Medicine.
 19. Yang B, Mallett S, Takwoingi Y, et al. QUADAS-C: a tool for assessing risk of bias in comparative diagnostic accuracy studies. *Ann Intern Med.* 2021;174(11):1592-1599. doi:10.7326/m21-2234
 20. Luyster FS, Strollo Jr. PJ, Zee PC, Walsh JK. Sleep: a health imperative. *Sleep.* 2012;35(6):727-734. doi:10.5665/sleep.1846
 21. Itani O, Jike M, Watanabe N, Kaneita Y. Short sleep duration and health outcomes: a systematic review, meta-analysis, and meta-regression. *Sleep Med.* 2017;32:246-256. doi:10.1016/j.sleep.2016.08.006
 22. Yaffe K, Falvey CM, Hoang T. Connections between sleep and cognition in older adults. *Lancet Neurol.* 2014;13(10):1017-1028. doi:10.1016/s1474-4422(14)70172-3
 23. Baglioni C, Battagliese G, Feige B, et al. Insomnia as a predictor of depression: a meta-analytic evaluation of longitudinal epidemiological studies. *J Affect Disord.* 2011;135(1-3):10-19. doi:10.1016/j.jad.2011.01.011
 24. Miller DJ, Sargent C, Roach GD. A validation of six wearable devices for estimating sleep, heart rate and heart rate variability in healthy adults. *Sensors.* 2022;22(16):6317. doi:10.3390/s22166317
 25. Lee T, Cho Y, Cha KS, et al. Accuracy of 11 wearable, nearable, and airable consumer sleep trackers: prospective multicenter validation study. *JMIR Mhealth Uhealth.* 2023;11:e50983. doi:10.2196/50983
 26. Nauha L, Farrahi V, Jurvelin H, et al. Comparison and agreement between device-estimated and self-reported sleep periods in adults. *Ann Med.* 2023;55(1):2191001. doi:10.1080/07853890.2023.2191001
 27. Redline S, Kirchner HL, Quan SF, Gottlieb DJ, Kapur V, Newman A. The effects of age, sex, ethnicity, and sleep-disordered breathing on sleep architecture. *Arch Intern Med.* 2004;164(4):406-418. doi:10.1001/archinte.164.4.406
 28. Pase MP, Harrison S, Misialek JR, et al. Sleep architecture, obstructive sleep apnea, and cognitive function in adults. *JAMA Netw Open.* 2023;6(7):e2325152. doi:10.1001/jamanetworkopen.2023.25152
 29. Pérez-Carbonell L, Muñoz-Lopetegi A, Sánchez-Valle R, et al. Sleep architecture and sleep-disordered breathing in fatal insomnia. *Sleep Med.* 2022;100:311-346. doi:10.1016/j.sleep.2022.08.027